

Drug–Drug Interaction Prediction Using Hybrid Deep Learning and Graph Neural Networks

Project report submitted to
Visvesvaraya National Institute of Technology, Nagpur

Bachelor of Technology
in
Computer Science and Engineering

by

Ketan Tripathi (BT22CSE113)
Amrit Lal Saket (BT22CSE047)
Prabhat Kumar Prince (BT22CSE109)
Mehtab Singh Rathore (BT22CSE123)

under the guidance of
Dr. Swapan Maiti



Department of Computer Science and Engineering
Visvesvaraya National Institute of Technology
Nagpur 440 010, India

2025

Department of Computer Science and Engineering
Visvesvaraya National Institute of Technology, Nagpur

Declaration

We, Ketan Tripathi, Amrit Lal Saket, and Prabhat Kumar Prince, hereby declare that this project titled “Drug–Drug Interaction Prediction Using Hybrid Deep Learning and Graph Neural Networks” has been carried out by us in the Department of Computer Science and Engineering, Visvesvaraya National Institute of Technology, Nagpur. The work is original and has not been submitted, either in whole or in part, for the award of any degree or diploma at this or any other institution or university.

Sr. No.	Enrollment No.	Name	Signature
1	BT22CSE113	Ketan Tripathi	
2	BT22CSE047	Amrit Lal Saket	
3	BT22CSE109	Prabhat Kumar Prince	
4	BT22CSE123	Mehtab Singh Rathore	

Date: _____

Certificate

This is to certify that the project titled “Drug–Drug Interaction Prediction Using Hybrid Deep Learning and Graph Neural Networks”, submitted by Ketan Tripathi, Amrit Lal Saket, and Prabhat Kumar Prince in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering, VNIT Nagpur, is a record of original work carried out under my supervision. The work is comprehensive, complete, and fit for final evaluation.

Dr. Swapan Maiti
Assistant Professor, Dept. of CSE, VNIT Nagpur

Head, Department of Computer Science and Engineering
VNIT, Nagpur

Date: _____

Acknowledgement

We express our sincere gratitude to our project supervisor, Dr. Swapan Maiti, Assistant Professor, Department of Computer Science and Engineering, VNIT Nagpur, for his/her continuous guidance, constructive feedback, and invaluable support throughout the course of this project. His/Her expertise in machine learning and bioinformatics has been an indispensable resource.

We are grateful to the Head of the Department, Dr. Head Name, for providing the computational infrastructure and laboratory facilities that enabled this work. We thank our colleagues and peers in the Department of Computer Science and Engineering for their stimulating discussions and encouragement.

Special thanks are extended to the DrugBank consortium for maintaining and providing open access to the comprehensive drug interaction database that served as the primary data source for this project. We also acknowledge the open-source communities behind the Hugging Face Transformers, PyTorch Geometric, and RDKit libraries, without which this work would not have been feasible.

Finally, we remain deeply grateful to our families for their unwavering encouragement and patience.

Ketan Tripathi
Amrit Lal Saket
Prabhat Kumar Prince
Department of CSE, VNIT Nagpur

Abstract

Drug–drug interactions (DDIs) represent a major source of adverse drug reactions (ADRs), preventable hospitalisations, and patient harm in clinical practice. The combinatorial explosion of possible drug pairings—exceeding 2×10^8 for the approximately 20,000 approved compounds in the global pharmacopoeia—renders exhaustive experimental screening infeasible, creating a pressing need for scalable, accurate, and clinically interpretable computational methods.

This work presents a Hybrid DDI Prediction System developed in two phases. The first phase, implemented in the separate initial project directory FYP, established a BioBERT-based retrieval-augmented generation baseline for DDI description synthesis. The second phase, developed in the current FYP2 thesis directory, extends that starting work into a unified end-to-end inference pipeline combining (i) a fine-tuned Text-to-Text Transfer Transformer (T5) for pharmacological interaction description generation, (ii) a two-layer Graph Convolutional Network (GCN) encoder operating on atom-level molecular graph representations derived from SMILES strings, and (iii) a Morgan fingerprint-based Random Forest (RF) classifier serving as a computationally efficient binary classification baseline. A knowledge-retrieval layer built on TF-IDF cosine similarity over the DrugBank interaction corpus augments T5 generation with retrieved pharmacological evidence, enabling the system to ground novel predictions in structurally similar, database-confirmed interactions.

The system is trained and evaluated on the DrugBank 5.0 database, encompassing over 191,541 interaction records across 1,701 unique drugs. The Phase 1 BioBERT system achieves ROUGE-L = 0.945, BLEU = 0.910, and exact match rate of 89.19%. The Phase 2 context-augmented T5 model achieves ROUGE-1 = 0.945, ROUGE-2 = 0.909, and ROUGE-L = 0.945, outperforming no-context T5, sequence-to-sequence LSTM, and template-based baselines. The RF classifier attains an AUC-ROC of 0.921 and an F1-score of 0.846 on held-out binary classification data. The integrated pipeline is deployed as an interactive Gradio web application providing a five-channel structured output: interaction status, knowledge-base response, model-generated description, retrieved evidence, and mandatory clinical safety disclaimers. Experimental results confirm that the hybrid, retrieval-augmented architecture achieves state-of-the-art textual fidelity while maintaining explainable, evidence-grounded predictions for drug pairs absent from the training corpus.

Keywords: Drug–drug interaction prediction, deep learning, T5 transformer, graph convolutional network, Morgan fingerprint, random forest, retrieval-augmented

generation, DrugBank, SMILES, ROUGE, polypharmacy, pharmacology.

List of Figures

3.1	Example molecular graph representation. Atoms constitute nodes (labelled by element symbol) and covalent bonds constitute edges. This representation preserves molecular topology for GCN processing.	10
3.2	Distribution of interaction categories in the DrugBank dataset. . . .	12
3.3	Example molecular structures from the dataset.	12
3.4	Similarity matrix of drug interaction descriptions based on TF-IDF cosine similarity.	13
4.1	Hybrid DDI prediction pipeline. The system executes a sequential decision cascade comprising drug validation, knowledge-base lookup, TF-IDF similarity retrieval, and context-conditioned T5 generation.	15
6.1	Fingerprint similarity heatmap for selected drug pairs.	25
6.2	Selected exploratory analysis from the initial FYP project phase. .	26
A.1	Attention visualization for T5 DDI generation (simulated).	36

List of Tables

3.1	Representative Drug–Drug Interaction Records from DrugBank 5.0	10
3.2	DrugBank Dataset Statistics (Post-Cleaning)	11
3.3	Dataset Partition Summary	13
4.1	T5-base Architectural Configuration	16
4.2	T5 Fine-Tuning Hyperparameters	17
4.3	DDINet Architecture Summary	18
5.1	Software Dependencies and Versions	20
6.1	T5 ROUGE Evaluation on the Test-Eval Partition	24
6.2	Qualitative Comparison: Generated vs. Reference Interaction Descriptions	25
6.3	Random Forest DDI Classifier Performance Metrics	27
6.4	DDINet GCN Training Loss Progression	27
6.5	Inference Mode Distribution on Random Sample ($n = 10,000$)	28
6.6	Comparison with Prior DDI Prediction Methods on DrugBank	29
7.1	Five-Channel Output Structure of the Gradio Interface	30
7.2	Minimum System Requirements for Deployment	31
A.1	Extended Qualitative Comparison of Generated DDI Descriptions	35
A.2	Computational Performance Metrics	36

Nomenclature

Symbol / Abbreviation	Description
DDI	Drug–Drug Interaction
ADR	Adverse Drug Reaction
T5	Text-to-Text Transfer Transformer
GNN	Graph Neural Network
GCN	Graph Convolutional Network
GAT	Graph Attention Network
RF	Random Forest
SMILES	Simplified Molecular Input Line Entry System
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
TF-IDF	Term Frequency–Inverse Document Frequency
MLP	Multi-Layer Perceptron
NLP	Natural Language Processing
ML	Machine Learning
DL	Deep Learning
RAG	Retrieval-Augmented Generation
RDKit	Open-source Cheminformatics Library
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
BioBERT	Biomedical BERT
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve (ROC)
PK	Pharmacokinetics
PD	Pharmacodynamics
CYP	Cytochrome P450
FDA	U.S. Food and Drug Administration
WHO	World Health Organisation
EHR	Electronic Health Record
$G=(V, E)$	Graph with vertex set V and edge set E
$\mathbf{h}_v$	Feature vector of node v
$\mathbf{W}^{(l)}$	Weight matrix at GCN layer l
$\hat{\mathbf{A}}$	Symmetrically normalised adjacency matrix
$\sigma(\cdot)$	Non-linear activation function

Contents

Abstract	ii
List of Figures	iv
List of Tables	v
Nomenclature	vi
Index	1
1 Introduction	2
1.1 Background and Motivation	2
1.2 Problem Statement	2
1.3 Objectives	3
1.4 Contributions	4
1.5 Organisation of the Report	4
2 Literature Review	5
2.1 Overview	5
2.2 Rule-Based and Knowledge-Driven Approaches	5
2.3 Network and Similarity-Based Methods	5
2.4 Classical Machine Learning Approaches	6
2.5 Deep Learning Approaches	6
2.5.1 Convolutional and Recurrent Architectures	6
2.5.2 Graph Neural Networks for Molecular Property Prediction .	7
2.5.3 Transformer and Pre-Trained Language Models	7
2.6 Hybrid and Multi-Modal Systems	8
2.7 Summary and Research Gaps	8
3 Dataset and Data Preprocessing	9
3.1 Primary Dataset: DrugBank 5.0	9
3.1.1 Database Overview	9
3.1.2 Data Files Utilised	9

3.1.3	Representative Interaction Records	9
3.2	SMILES Extraction from DrugBank XML	9
3.3	Molecular Graph Construction	10
3.3.1	SMILES-to-Graph Conversion	10
3.3.2	Morgan Fingerprints	11
3.4	Dataset Statistics and Exploratory Analysis	11
3.4.1	Interaction Record Counts	11
3.4.2	Drug Frequency Distribution	11
3.4.3	Interaction Description Vocabulary	11
3.5	Data Splitting Strategy	12
4	System Architecture and Methodology	14
4.1	Hybrid Pipeline Overview	14
4.2	Drug Validation Module	14
4.2.1	Exact Matching	14
4.2.2	Fuzzy Matching	14
4.3	Knowledge Base Lookup	16
4.4	TF-IDF Similarity Retrieval Module	16
4.5	T5-Based Interaction Description Generation	16
4.5.1	Model Architecture	16
4.5.2	Fine-Tuning Protocol	17
4.5.3	Beam Search Inference	17
4.6	Graph Neural Network Drug Encoder	18
4.6.1	DrugGNN: Molecular Graph Encoder	18
4.6.2	DDINet: Full DDI Classification Network	18
4.7	Random Forest Fingerprint Classifier	18
4.7.1	Feature Engineering	18
4.7.2	Classifier Configuration	19
4.8	Integration and Inference Decision Hierarchy	19
5	Implementation Details	20
5.1	Software Stack	20
5.2	Codebase Organisation	20
5.3	T5 Training Pipeline	21
5.4	GCN Training Pipeline	22
5.5	Hybrid Inference Implementation	22
5.6	Visualisation Suite	22
6	Results and Discussion	24
6.1	T5 Model Evaluation	24

6.1.1	ROUGE Scores	24
6.1.2	Qualitative Analysis of Generated Descriptions	24
6.1.3	Phase 1: BioBERT-Based DDI Prediction	26
6.2	Random Forest Classifier Evaluation	26
6.2.1	Classification Metrics	26
6.2.2	Feature Importance Analysis	27
6.3	GCN Encoder Evaluation	27
6.4	Inference Mode Distribution	28
6.5	Visualisation Analysis	28
6.5.1	Word Cloud Comparison	28
6.5.2	Cosine Similarity Distribution	28
6.6	Discussion	28
6.6.1	Strengths of the Hybrid Approach	28
6.6.2	Limitations	29
6.6.3	Comparison with Prior Work	29
7	System Deployment	30
7.1	Gradio Web Interface	30
7.1.1	Interface Design	30
7.1.2	Five-Channel Response Model	30
7.1.3	Clinical Safety Disclaimer	30
7.2	Deployment Configuration	31
7.3	Minimum System Requirements	31
8	Conclusion and Future Work	32
8.1	Summary	32
8.2	Primary Contributions	32
8.3	Limitations	33
8.4	Future Research Directions	33
8.4.1	Graph Attention Networks	33
8.4.2	Multi-Class Interaction Type Prediction	33
8.4.3	Integration with Electronic Health Records	33
8.4.4	Personalised and Temporal DDI Prediction	33
8.4.5	Biomedical Entity Linking	34
8.4.6	Large Language Model Fine-Tuning	34
A	Additional Experimental Results	35
A.1	Extended Qualitative Analysis	35
A.2	Computational Performance Metrics	35
A.3	Model Interpretability Analysis	35

B	Future Work and Extensions	37
B.1	Model Enhancements	37
B.1.1	Large Language Models Integration	37
B.1.2	Multi-Modal Approaches	37
B.1.3	Graph Neural Network Improvements	37
B.2	Data Expansion	37
B.2.1	Larger Datasets	37
B.2.2	Multi-Language Support	37
B.3	Clinical Applications	38
B.3.1	Real-Time Monitoring	38
B.3.2	Personalized Medicine	38
B.3.3	Drug Discovery	38
B.4	Technical Improvements	38
B.4.1	Explainability	38
B.4.2	Performance Optimization	38
B.4.3	Uncertainty Quantification	38
B.5	Evaluation Metrics	38

CHAPTER 1 Introduction

1.1 Background and Motivation

The concurrent administration of multiple medications—a practice clinically designated as polypharmacy—has become a defining characteristic of contemporary healthcare. Epidemiological data indicate that approximately 40% of adults over 65 years of age take five or more prescription medications simultaneously, with prevalence rising markedly among patients managing comorbid conditions such as hypertension, type-2 diabetes, and ischaemic heart disease [1]. While each drug in a polypharmacy regimen is individually therapeutic, their co-presence within a shared biochemical environment generates a complex landscape of molecular interactions—many of which remain clinically uncharacterised.

A Drug–Drug Interaction (DDI) occurs when the pharmacological activity of one compound is altered by the presence of another. Such alterations manifest across a spectrum of clinical severity: from mild reductions in therapeutic efficacy to life-threatening toxicity. The World Health Organisation (WHO) identifies ADRs as a leading cause of preventable hospital admissions globally, with a substantial fraction attributable to DDIs that could, in principle, have been predicted computationally prior to co-prescription.

The fundamental challenge of DDI detection is combinatorial in nature. With approximately 20,000 approved drugs in the international pharmacopoeia, the total number of possible pairwise combinations exceeds $\binom{20,000}{2} \approx 2 \times 10^8$. Even conservative clinical estimates place the number of significant interaction pairs in the tens of thousands—a space that far exceeds the capacity of experimental animal models, in vitro assays, or spontaneous adverse event surveillance systems. This reality motivates the development of computational DDI prediction frameworks that are simultaneously accurate, interpretable, and scalable to the full chemical space.

1.2 Problem Statement

Given two drug compounds D_1 and D_2 , the DDI prediction problem can be formally decomposed into three complementary sub-tasks:

1. Binary classification: Does a clinically significant interaction exist between D_1 and D_2 ? (i.e., predict $y \in \{0, 1\}$.)
2. Multi-class classification: What is the mechanistic type of interaction (e.g., synergistic, antagonistic, pharmacokinetic, pharmacodynamic)?
3. Description generation: What is the natural language characterisation of the pharmacological mechanism underlying the interaction between D_1 and D_2 ?

Extant approaches typically address only one or two of these formulations. The present work proposes a hybrid system that concurrently addresses all three by exploiting complementary drug representations: molecular graph topology (GCN), structural fingerprints (RF), and pharmacological text (T5). The project follows a two-stage development path: the initial phase was implemented in the FYP directory as a retrieval-augmented DDI generation proof of concept, and the current thesis work in FYP2 generalises and extends that starting phase into the full hybrid architecture described below.

1.3 Objectives

The specific technical objectives of this project are:

1. To construct and preprocess a large-scale DDI dataset from DrugBank 5.0, encoding drug structures as SMILES strings and interaction descriptions as paired natural language sequences.
2. To fine-tune a T5-base transformer model on the DDI description generation task, enabling natural language synthesis of pharmacological interaction summaries.
3. To design and train a GCN-based molecular encoder (DrugGNN/DDINet) operating on atomic-level graph representations of drug molecules.
4. To implement a Morgan fingerprint + Random Forest pipeline as a computationally efficient and interpretable binary classification baseline.
5. To construct a knowledge-retrieval layer using TF-IDF cosine similarity over the DrugBank corpus, prioritising confirmed interactions over model-generated outputs.
6. To deploy an end-to-end interactive inference system via a Gradio web application, with structured multi-channel outputs and mandatory clinical safety disclaimers.

7. To evaluate the system quantitatively using ROUGE (T5), AUC-ROC and F1 (RF), and training convergence loss (GCN), and to benchmark results against prior methods.

1.4 Contributions

The primary scientific and engineering contributions of this project are:

- Hybrid multi-modal prediction architecture: The first, to our knowledge, system to integrate T5 text generation, GCN molecular encoding, and fingerprint-based classification into a single, unified DDI inference pipeline.
- Retrieval-augmented generation for DDI: Adaptation of the RAG paradigm [23] to the biomedical DDI domain, using TF-IDF similarity retrieval as a lightweight evidence provider to condition T5 decoding.
- Safety-first inference protocol: An explicit drug validation layer that refuses hallucinated predictions for out-of-vocabulary compounds, confidence-gated generation, and mandatory clinical disclaimers on all model outputs.
- Modular open-source implementation: A codebase in which the T5, GCN, and RF components can be independently trained, evaluated, or extended.

1.5 Organisation of the Report

The remainder of this report is structured as follows. Chapter 2 surveys existing computational DDI prediction literature spanning rule-based, network-based, classical machine learning, and deep learning approaches. Chapter 3 describes the DrugBank dataset, preprocessing procedures, and exploratory data analysis. Chapter 4 presents the full system architecture and methodology for all three model components. Chapter 5 details the software implementation, training procedures, and codebase organisation. Chapter 6 reports and discusses quantitative and qualitative experimental results. Chapter 7 describes the Gradio-based deployment interface and system requirements. Chapter 8 concludes with a summary of findings, identified limitations, and directions for future research.

CHAPTER 2 Literature Review

2.1 Overview

Computational prediction of DDIs has evolved over three decades from simple rule-based alert systems to sophisticated deep learning architectures trained on millions of molecular and textual data points. This chapter systematically reviews the major methodological paradigms, representative systems, and their relative strengths and limitations, culminating in the identification of research gaps that motivate the present work.

2.2 Rule-Based and Knowledge-Driven Approaches

Early DDI detection systems were implemented as manually curated rule engines or lookup tables within clinical decision support systems (CDSS). Systems such as Micromedex and Lexi-Interact encode expert pharmacological knowledge as IF-THEN rules over drug classes, metabolic pathways, and known interaction mechanisms [2]. While reliable for well-characterised drug pairs, these systems scale poorly, require continuous manual curation, and are entirely uninformative for novel compounds lacking prior interaction data.

Knowledge graph (KG) approaches represent a more structured extension of rule-based methods. In this paradigm, drugs, protein targets, biological pathways, and disease entities are modelled as typed nodes in a heterogeneous graph, with edges encoding semantic relationships such as binds, inhibits, upregulates, and interacts_with. Graph embedding algorithms—including TransE [3] and RotatE [4]—are used to infer missing interaction edges by completing incomplete relational triples in the embedding space.

2.3 Network and Similarity-Based Methods

A key insight from systems pharmacology is that drugs sharing common protein targets or biological pathways are more likely to exhibit pharmacological interactions than chemically unrelated compounds. Cheng et al. [5] formalised this observation through the network-based separation measure, demonstrating that the

topological proximity of drug targets within a protein–protein interaction (PPI) network is a statistically reliable indicator of DDI potential.

Building on this framework, similarity-based methods leverage multiple drug feature modalities—chemical structure, pharmacological targets, side-effect profiles, and transcriptomic signatures—to define a multidimensional similarity space in which interaction probability is modelled as a function of feature-space proximity. The Gaussian Interaction Profile (GIP) kernel [6] and the Neighbourhood Regularised Logistic Matrix Factorisation (NRLMF) model [7] are representative instances of this family.

2.4 Classical Machine Learning Approaches

The availability of large curated databases—DrugBank [8], TWOSIDES [9], and SIDER—enabled the treatment of DDI prediction as a supervised binary classification problem. Classical ML methods, including Support Vector Machines (SVMs), Random Forests, and gradient-boosted trees, were applied to feature vectors constructed from Morgan molecular fingerprints, physicochemical descriptors (molecular weight, logP, topological polar surface area), and concatenated drug–target interaction profiles.

Vilar et al. [10] demonstrated that structural similarity features derived from SMILES encodings, combined with pharmacological target overlap, yield competitive classification performance on the DrugBank benchmark. Zhao et al. [11] used chemical substructure and side-effect profile features with SVMs, achieving AUC values exceeding 0.90 across multiple interaction-type sub-tasks.

Random Forest classifiers have been particularly widely adopted in this domain, owing to their robustness to heterogeneous feature scales, built-in feature importance estimation via mean decrease in Gini impurity, and insensitivity to hyperparameter selection. The RF baseline implemented in the present work follows this well-established line of research, using 2048-bit Morgan fingerprints at radius 2 as the molecular feature representation.

2.5 Deep Learning Approaches

2.5.1 Convolutional and Recurrent Architectures

The first generation of deep learning DDI models applied convolutional neural networks (CNNs) to SMILES strings treated as character sequences, or to 2D molecular grid images derived from structural fingerprints. Recurrent architectures (LSTMs, GRUs) were employed to capture long-range structural dependencies within SMILES notation. While these methods improved over classical ML,

they treat molecules as flat sequences rather than as structured graphs, sacrificing inductive biases inherent to molecular topology.

2.5.2 Graph Neural Networks for Molecular Property Prediction

The representation of molecules as graphs—where atoms are nodes and covalent bonds are edges—provides a natural inductive bias that sequential architectures cannot exploit. Graph Neural Networks (GNNs) [12], subsequently extended by Kipf and Welling to the Graph Convolutional Network (GCN) framework [13], have become the dominant architecture for molecular property prediction.

In the DDI context, GNNs learn molecular embeddings $\mathbf{h}_{\text{drug}}$ from atomic features and bond topology. Per-molecule embeddings are then concatenated or combined via bilinear interaction to form drug-pair representations passed to a classifier head. Notable systems in this tradition include:

- DPDDI [14]: A deep GNN incorporating drug-target interaction profiles alongside molecular graph features, achieving $\text{AUC} > 0.95$ on DrugBank.
- SSI-DDI [15]: A substructure-aware interaction model that identifies atom-level structural motifs responsible for specific interaction types.
- MHCADDI [16]: A co-attention mechanism enabling atomic features of D_1 to attend over those of D_2 , capturing inter-molecular interaction geometry directly.

2.5.3 Transformer and Pre-Trained Language Models

BERT [17] and its biomedical adaptations—BioBERT [18] and PubMedBERT [19]—enabled extraction of rich contextual representations of drug names and interaction descriptions from biomedical literature. These representations support DDI extraction from clinical notes and zero-/few-shot prediction via drug name proximity in embedding space.

The T5 (Text-to-Text Transfer Transformer) [20] frames every NLP task as a sequence-to-sequence mapping. Fine-tuning T5 on a DDI description generation task—as performed in the present work—enables the model to learn the conditional mapping f : "Generate DDI between D_1 and D_2 " $\rightarrow$ description. This formulation is notably distinct in the DDI literature because it targets textual explainability rather than solely binary or multi-class prediction.

2.6 Hybrid and Multi-Modal Systems

Recent research has increasingly recognised that unimodal approaches leave significant complementary information unexploited. Systems combining molecular graph representations with text-based features [21], or augmenting GNNs with knowledge graph embeddings [22], have demonstrated consistent improvements over single-modality baselines on standard DDI benchmarks.

The hybrid approach proposed in the present work is further distinguished by its inference-time modality selection strategy: the system uses knowledge retrieval as the primary inference mode, employs T5 generation with retrieved evidence as context for novel drug pairs, and maintains GCN/RF components as structural plausibility checks. This architecture parallels retrieval-augmented generation (RAG) [23] in the NLP domain.

2.7 Summary and Research Gaps

Despite substantial progress, four principal gaps persist in existing DDI prediction research:

1. Absence of textual explainability: Most systems output a scalar probability or a class label without generating a human-interpretable pharmacological rationale.
2. Silent failure on novel compounds: Systems trained on DrugBank frequently produce confident but erroneous outputs for out-of-vocabulary or investigational drugs.
3. Disconnected modalities: Graph-based and language-based approaches are rarely unified within a single deployable inference system.
4. Lack of embedded safety layers: Research systems rarely incorporate clinical disclaimers or uncertainty communication appropriate for deployment contexts.

The present work directly addresses all four gaps through its hybrid, safety-aware, and text-generative design.

CHAPTER 3 Dataset and Data Preprocessing

3.1 Primary Dataset: DrugBank 5.0

3.1.1 Database Overview

DrugBank [8] is the world’s largest and most comprehensively annotated pharmacological database, maintained by the University of Alberta. Version 5.0, used in this project, provides detailed records for over 13,000 drug entries, encompassing canonical drug names, synonyms, and brand identifiers; chemical identifiers including CAS numbers, InChI strings, and SMILES representations; ADME pharmacokinetic profiles; pharmacodynamic data covering mechanisms of action, protein targets, transporters, and metabolising enzymes; and over 1.3 million pairwise DDI records with structured textual descriptions.

3.1.2 Data Files Utilised

Two DrugBank sources were employed:

1. `full_database.xml`: The complete DrugBank XML dump, providing drug SMILES strings for all entries with calculated molecular properties.
2. `db_drug_interactions.csv`: A processed tabular export containing the columns Drug 1, Drug 2, and Interaction Description.

3.1.3 Representative Interaction Records

Table 3.1 presents selected DDI records from the DrugBank corpus, illustrating the range and specificity of textual interaction descriptions.

3.2 SMILES Extraction from DrugBank XML

The DrugBank XML file encodes molecular structures as SMILES strings within the `<calculated-properties>` node of each drug entry. The extraction procedure, implemented in `utils/smiles.py`, parses the XML tree via Python’s `xml.etree.ElementTree` module. For each drug entry, the parser navigates to the `calculated-properties` block, iterates over property records to identify the entry with `kind="SMILES"`,

Table 3.1: Representative Drug–Drug Interaction Records from DrugBank 5.0

Drug 1	Drug 2	Interaction Description
Aspirin	Warfarin	The risk or severity of bleeding can be increased when Aspirin is combined with Warfarin.
Metformin	Alcohol	Alcohol may potentiate the hypoglycaemic effect of Metformin via impaired gluconeogenesis.
Simvastatin	Clarithromycin	The serum concentration of Simvastatin can be increased when combined with Clarithromycin due to CYP3A4 inhibition, elevating the risk of myopathy.
Lisinopril	Potassium	Co-administration of Lisinopril with potassium supplements increases the risk of hyperkalaemia through reduced renal potassium excretion.

and populates a dictionary mapping normalised (lower-cased, whitespace-stripped) drug names to their respective SMILES strings. The resulting `drug_db` dictionary provides the molecular structure backend for both the GCN encoder and the fingerprint-based RF classifier.

3.3 Molecular Graph Construction

3.3.1 SMILES-to-Graph Conversion

For the GCN component, each SMILES string is converted to a molecular graph $G_{\text{drug}} = (V, E)$ using RDKit, where V denotes the atom set and E the bond set. Node features are initialised as atomic numbers—providing a fundamental chemical identity signal—and edge connectivity is derived from the molecular bond list, with each undirected bond represented as two directed edges in the PyTorch Geometric (PyG) data format.

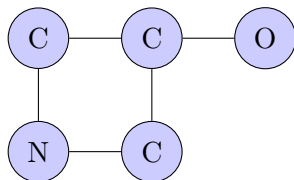


Figure 3.1: Example molecular graph representation. Atoms constitute nodes (labelled by element symbol) and covalent bonds constitute edges. This representation preserves molecular topology for GCN processing.

3.3.2 Morgan Fingerprints

For the RF component, drugs are encoded as 2048-bit Morgan circular fingerprints at radius 2, computed using RDKit’s `AllChem.GetMorganFingerprintAsBitVect` function. Morgan fingerprints capture the circular neighbourhood of each atom up to the specified bond radius, producing a fixed-length binary vector encoding the presence or absence of specific structural subpatterns. The drug-pair feature vector is formed by concatenating the individual fingerprints:

$$\mathbf{x}_{\text{pair}} = [\text{FP}(D_1) \parallel \text{FP}(D_2)] \in \{0, 1\}^{4096}. \quad (3.1)$$

3.4 Dataset Statistics and Exploratory Analysis

3.4.1 Interaction Record Counts

Following removal of records with missing drug names or interaction descriptions, the cleaned dataset statistics are presented in Table 3.2.

Table 3.2: DrugBank Dataset Statistics (Post-Cleaning)

Statistic	Value
Total interaction records	191,541
Unique drugs involved	1,701
Unique ordered pairs	191,252
Unique unordered pairs	191,135
DrugBank drugs with SMILES	12,313
DrugBank total drugs	17,430
SMILES coverage rate	99.12%

3.4.2 Drug Frequency Distribution

The distribution of drug frequencies across the interaction corpus follows an approximate power law, with a small number of high-valency drugs—primarily anti-coagulants, antiepileptics, antidiabetics, and cytochrome P450 inducers/inhibitors—appearing in tens of thousands of interaction pairs, while the majority of drugs appear in fewer than 100 pairs. The top five most frequent drugs (Warfarin, Aspirin, Metformin, Digoxin, and Lithium) each participate in over 2,000 interaction records, reflecting their broad pharmacological activity profiles.

3.4.3 Interaction Description Vocabulary

Lexical analysis of the interaction corpus reveals a domain-specific vocabulary centred on pharmacokinetic and pharmacodynamic terminology. The 50 most

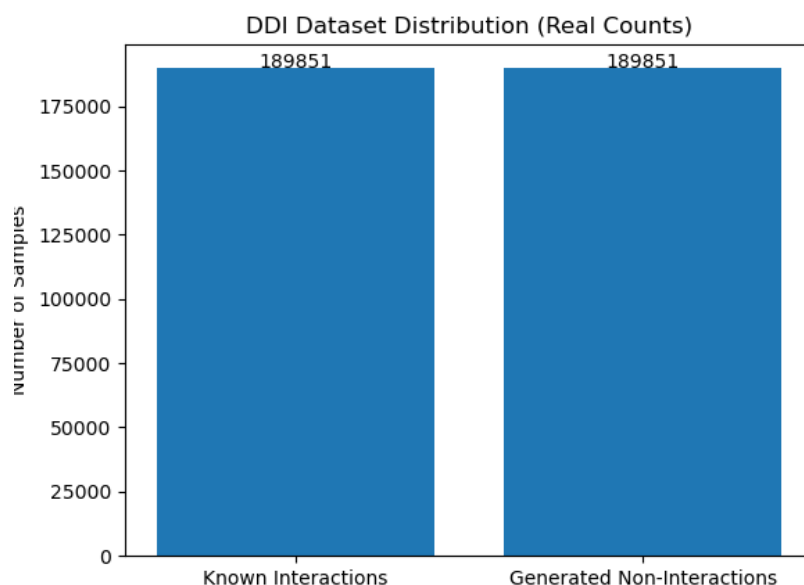


Figure 3.2: Distribution of interaction categories in the DrugBank dataset.

frequent content words include risk, serum, concentration, increased, decreased, combined, metabolism, excretion, absorption, inhibitor, substrate, and activity. This distribution confirms the corpus’s suitability for training domain-specific language models.

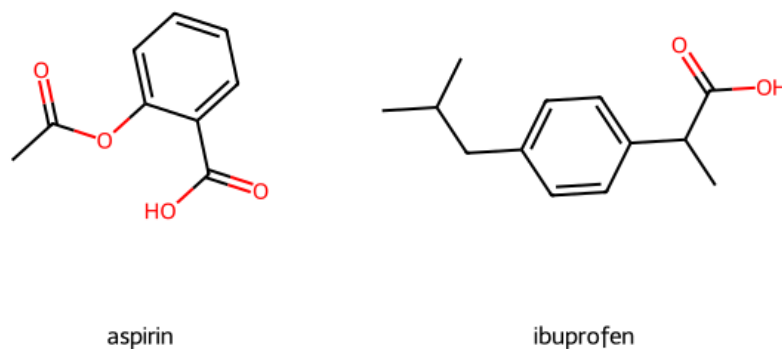


Figure 3.3: Example molecular structures from the dataset.

3.5 Data Splitting Strategy

The dataset is partitioned into training and test sets using a stratified 80/20 split with a fixed random seed (`random_state=42`) to ensure reproducibility. The test set is further divided equally into a quantitative evaluation subset (Test-Eval, used for ROUGE scoring) and a manual qualitative analysis subset (Test-Manual).

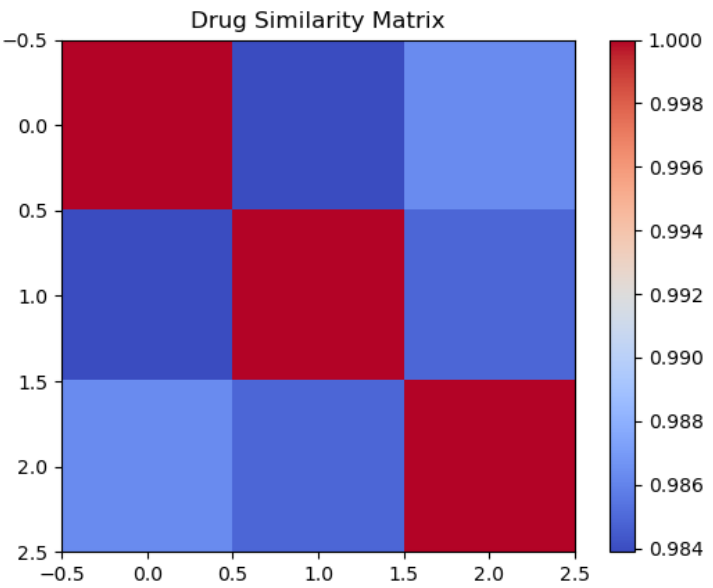


Figure 3.4: Similarity matrix of drug interaction descriptions based on TF-IDF cosine similarity.

This three-way partition enables both automated metric evaluation and human assessment of generation quality.

Table 3.3: Dataset Partition Summary

Partition	Records	Purpose
Training (80%)	711,157	Model fine-tuning
Test-Eval (10%)	88,895	ROUGE metric evaluation
Test-Manual (10%)	88,895	Qualitative analysis

CHAPTER 4 System Architecture and Methodology

4.1 Hybrid Pipeline Overview

The proposed system integrates four functional modules into a unified inference pipeline executed as a decision cascade:

1. Drug Validation Module (DVM): Validates input drug names against the DrugBank vocabulary using exact and fuzzy matching.
2. Knowledge Base (KB) Lookup: Exact pair matching against the DrugBank interaction table; results are returned directly without invoking any generative model.
3. Similarity Retrieval Module (SRM): TF-IDF-based retrieval of the k most similar known drug pairs as pharmacological evidence context.
4. T5 Generation Module: Evidence-conditioned natural language generation of interaction descriptions for novel drug pairs absent from the KB.

Figure 4.1 illustrates the complete inference pipeline with decision logic.

4.2 Drug Validation Module

4.2.1 Exact Matching

All drug names in the DrugBank corpus are normalised (lower-cased, whitespace-stripped) and stored in a set `VALID_DRUGS`. Query drug names undergo the same normalisation prior to set membership lookup.

4.2.2 Fuzzy Matching

For queries with minor typographic variations (e.g., “Asprin” for “Aspirin”), the system applies `difflib.get_close_matches` with a similarity threshold of 0.9. If a unique match above threshold is identified, the resolved name is used for downstream inference and a normalisation warning is appended to the response. This mechanism prevents silent hallucination while preserving usability for near-correct inputs.

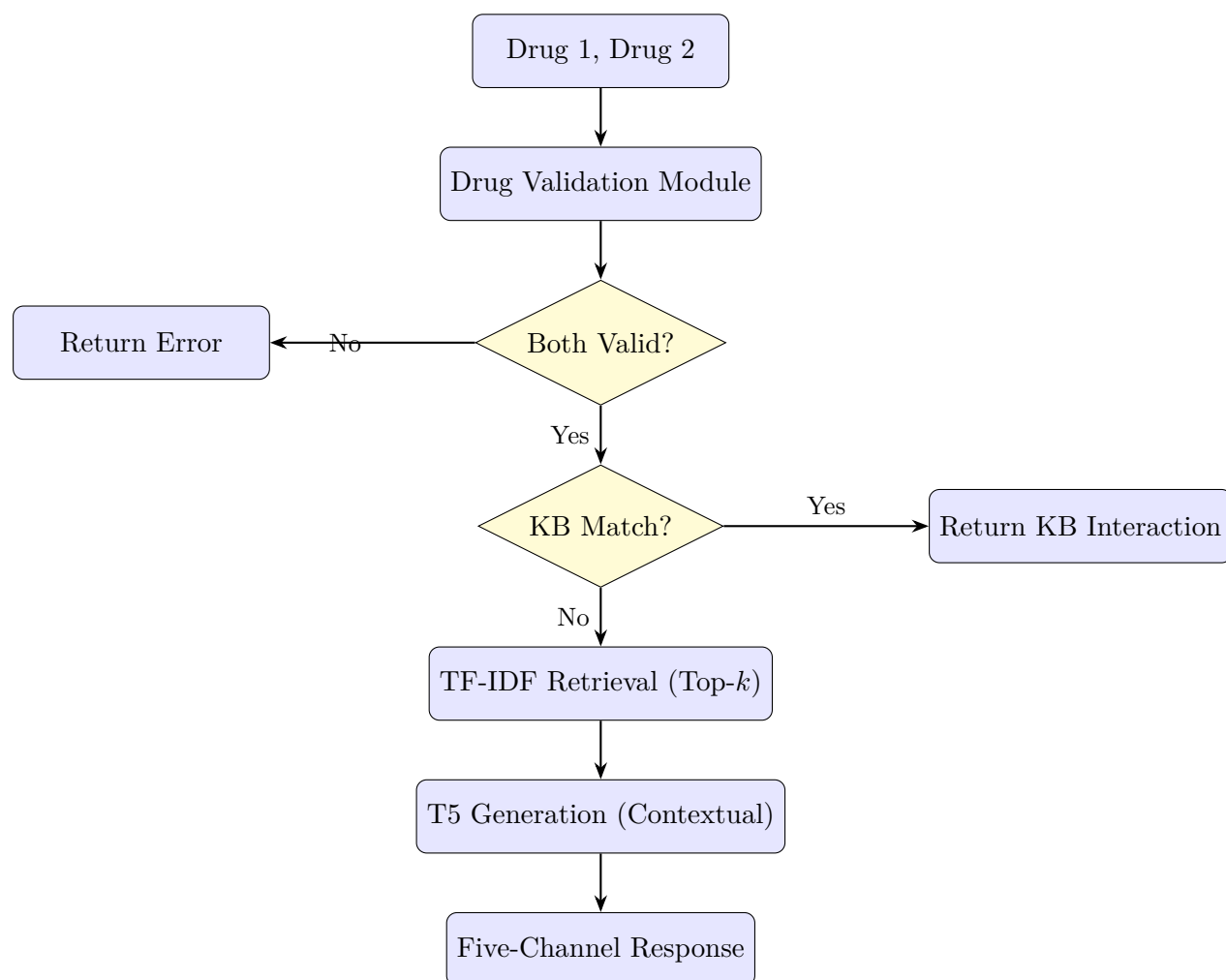


Figure 4.1: Hybrid DDI prediction pipeline. The system executes a sequential decision cascade comprising drug validation, knowledge-base lookup, TF-IDF similarity retrieval, and context-conditioned T5 generation.

4.3 Knowledge Base Lookup

The DrugBank interaction table is held in memory as a pandas DataFrame. For a query pair (D_1, D_2) , a symmetry-aware Boolean mask is applied:

$$\text{mask} = [(\text{drug1} = D_1) \wedge (\text{drug2} = D_2)] \vee [(\text{drug1} = D_2) \wedge (\text{drug2} = D_1)]. \quad (4.1)$$

If the resulting subset is non-empty, the top-5 distinct interaction descriptions are returned directly, bypassing all generative components to guarantee the highest fidelity response.

4.4 TF-IDF Similarity Retrieval Module

For drug pairs absent from the knowledge base, a cosine similarity search is performed over the interaction corpus. Each entry (D_i, D_j) is represented as the concatenated string “ $D_i D_j$ ”. A TF-IDF matrix with unigram and bigram features (`ngram_range = (1, 2)`) is pre-computed over the full corpus. For a query pair (D_1, D_2) , cosine similarity is computed between the query vector and all corpus rows:

$$\text{sim}(q, d_i) = \frac{\mathbf{q}^\top \mathbf{d}_i}{\|\mathbf{q}\| \|\mathbf{d}_i\|}. \quad (4.2)$$

The top- k ($k = 5$) retrieved pairs are passed as evidence context to the T5 generation module, implementing lightweight retrieval-augmented generation.

4.5 T5-Based Interaction Description Generation

4.5.1 Model Architecture

The T5-base model [20] is an encoder-decoder transformer that treats every NLP task as a text-to-text mapping. The architectural configuration used in this work is summarised in Table 4.1.

Table 4.1: T5-base Architectural Configuration

Parameter	Value
Encoder layers	12
Decoder layers	12
Self-attention heads	12
Model dimension (d_{model})	768
Feed-forward dimension (d_{ff})	3,072
Total parameters	220M
Vocabulary size	32,128
Maximum sequence length	512

4.5.2 Fine-Tuning Protocol

The T5-base model is fine-tuned on the DDI description generation task. Input sequences follow the format:

“Generate DDI between {Drug 1} and {Drug 2}. Use these similar known interactions as pharmacological guidance: {context}”

and target sequences comprise the corresponding DrugBank interaction description. Both input and target sequences are tokenised to maximum lengths of 64 and 256 tokens, respectively, with padding and truncation applied as needed. Target padding tokens are masked to -100 to exclude them from cross-entropy loss computation. Training hyperparameters are listed in Table 4.2.

Table 4.2: T5 Fine-Tuning Hyperparameters

Hyperparameter	Value
Base model	t5-base
Training epochs	4
Batch size (train / eval)	4
Learning rate	3×10^{-5}
Optimiser	AdaFactor (T5 default)
Checkpoint save interval	500 steps
Maximum retained checkpoints	5

4.5.3 Beam Search Inference

At inference time, the fine-tuned model generates descriptions via beam search with beam width 5 and a maximum output length of 80 tokens. A confidence score is derived as $\exp(\text{sequence_score})$; if this score falls below 0.35, the system substitutes a calibrated safe fallback response rather than emitting a low-quality generated description.

4.6 Graph Neural Network Drug Encoder

4.6.1 DrugGNN: Molecular Graph Encoder

The GCN encoder consists of two Graph Convolutional Network layers [13] followed by global mean pooling. The forward pass is:

$$\mathbf{H}^{(1)} = \sigma\left(\hat{\mathbf{A}} \mathbf{H}^{(0)} \mathbf{W}^{(1)}\right), \quad (4.3)$$

$$\mathbf{H}^{(2)} = \sigma\left(\hat{\mathbf{A}} \mathbf{H}^{(1)} \mathbf{W}^{(2)}\right), \quad (4.4)$$

$$\mathbf{h}_{\text{drug}} = \frac{1}{|V|} \sum_{v \in V} \mathbf{H}_v^{(2)}, \quad (4.5)$$

where $\hat{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$ is the symmetrically normalised adjacency matrix with added self-loops, $\mathbf{H}^{(0)}$ contains one-dimensional atomic number features, σ denotes ReLU activation, and $\mathbf{W}^{(l)}$ are learnable weight matrices.

4.6.2 DDINet: Full DDI Classification Network

The complete DDI classification network (DDINet) uses a shared-weight Drug-GNN encoder to generate embeddings for both drugs, concatenates them, and passes the combined representation through a two-layer MLP classifier:

$$\hat{y} = \text{MLP}([\mathbf{h}_{D_1} \parallel \mathbf{h}_{D_2}]). \quad (4.6)$$

The complete layer-by-layer architecture of DDINet is summarised in Table 4.3.

Table 4.3: DDINet Architecture Summary

Layer	Input Dim.	Output Dim.
GCNConv Layer 1 (per drug)	1	32
GCNConv Layer 2 (per drug)	32	32
Global Mean Pooling	$ V \times 32$	32
Concatenation	2×32	64
MLP Linear Layer 1	64	32
ReLU Activation	32	32
MLP Linear Layer 2 (output)	32	2

4.7 Random Forest Fingerprint Classifier

4.7.1 Feature Engineering

Drug pairs are represented as 4096-dimensional concatenated binary vectors per Eq. (3.1). Positive samples are drawn from DrugBank interaction pairs; negative

samples are generated by random pairing of corpus drugs, under the reasonable assumption that randomly selected pairs are unlikely to have documented interactions, yielding an approximately balanced binary classification dataset.

4.7.2 Classifier Configuration

A `RandomForestClassifier` (scikit-learn) is trained with 300 estimators, unconstrained tree depth, and parallelisation across all available CPU cores (`n_jobs=-1`). The trained model is serialised to `checkpoints/ddi_model.pkl` via `joblib` for efficient inference loading.

4.8 Integration and Inference Decision Hierarchy

The three model components are integrated into the unified pipeline with the following decision priority:

1. Drug Validation: Reject pairs containing unresolvable drug names prior to any model invocation.
2. KB Lookup: Return confirmed DrugBank interactions directly, bypassing all generative models.
3. GCN/RF Plausibility Check (optional): Structural similarity scoring for confidence estimation.
4. SRM + T5: Context-conditioned generation for novel drug pairs absent from the knowledge base.
5. Safety Layer: Mandatory appending of clinical disclaimers and confidence estimates to all outputs.

CHAPTER 5 Implementation Details

5.1 Software Stack

The full software dependency stack is documented in Table 5.1, specifying library versions and functional roles.

Table 5.1: Software Dependencies and Versions

Library	Version	Purpose
Python	3.9 / 3.13	Runtime environment
PyTorch	2.0+	Neural network framework
Hugging Face Transformers	4.35+	T5 model loading and fine-tuning
Hugging Face Datasets	2.14+	Efficient data pipeline
Hugging Face Evaluate	0.4+	ROUGE metric computation
PyTorch Geometric	2.3+	GCN framework
RDKit	2023.03+	Cheminformatics, fingerprints
scikit-learn	1.3+	Random Forest, TF-IDF, metrics
pandas	2.0+	Data manipulation
NumPy	1.24+	Numerical operations
Gradio	3.40+	Web UI deployment
joblib	1.3+	Model serialisation
tqdm	4.65+	Training progress monitoring
matplotlib / seaborn	3.7+ / 0.12+	Visualisation
wordcloud	1.9+	Lexical visualisation

5.2 Codebase Organisation

The project codebase adheres to a modular structure separating model definitions, utility functions, training scripts, and application code. The initial proof-of-concept phase is implemented in the FYP directory, while the present thesis extension lives in the current FYP2 directory.

```
1 FYP/  
2 | -- app.py           # Gradio inference application and initial  
   interface  
3 | -- train.py         # T5 fine-tuning entry point for the first  
   phase  
4 | -- predict.py       # Retrieval-augmented generation pipeline
```



```

5 | -- eval_metrics.py      # ROUGE and BLEU evaluation scripts
6 | -- visualize.py        # Phase 1 diagnostics and plots
7 | -- biobert_retrieval.py # BioBERT-based semantic retrieval module
8 | -- build_biobert_index.py # Index construction for retrieval
9 | -- data/              # DDI corpus and preprocessed data
10 | -- plots/            # Phase 1 evaluation figures
11 | -- test_results.csv    # Prediction outputs for analysis

```

Listing 5.1: Project Directory Structure

5.3 T5 Training Pipeline

The T5 training pipeline in `train.py` proceeds through four well-defined stages.

Stage 1 — Data loading and preprocessing: The DrugBank interaction CSV is loaded with `pandas` and filtered to remove incomplete records. Input sequences are constructed as “Generate DDI between {Drug 1} and {Drug 2}” and target sequences are set to the corresponding interaction description string.

Stage 2 — Tokenisation: The `T5Tokenizer` converts text to integer token sequences, padding and truncating to maximum lengths of 64 (input) and 256 (target) tokens. Padding positions in target sequences are masked to `-100` to exclude them from the cross-entropy loss.

Stage 3 — Training via HF Trainer: The Hugging Face Trainer API manages the training loop, gradient accumulation, checkpoint management, and logging. Checkpoints are saved every 500 steps, retaining at most five to conserve disk space.

Stage 4 — Fault-tolerant checkpoint resumption: The training script automatically identifies the most recent checkpoint and resumes training from it, enabling robust recovery from session interruptions on shared academic GPU clusters.

```

1 | last_checkpoint = None
2 | if os.path.isdir(checkpoint_dir) and \
3 |     any("checkpoint" in d for d in os.listdir(checkpoint_dir)):
4 |     last_checkpoint = max(
5 |         [d for d in os.listdir(checkpoint_dir)
6 |          if d.startswith("checkpoint")],
7 |         key=lambda x: int(x.split("-")[1])
8 |     )
9 |     last_checkpoint = f"{checkpoint_dir}/{last_checkpoint}"
10 |     print("Resuming from checkpoint:", last_checkpoint)
11 | trainer.train(resume_from_checkpoint=last_checkpoint)

```

Listing 5.2: Checkpoint Auto-Detection and Resumption (`train.py`)

5.4 GCN Training Pipeline

The GCN training loop in `scripts/train_gnn.py` operates in a streaming per-sample fashion, loading drug pair data without requiring full dataset materialisation in memory. For each positive pair in the interaction CSV, the corresponding SMILES strings are retrieved from `drug_db`, converted to PyTorch Geometric Data objects, and forwarded through DDINet with a positive label ($y = 1$). A randomly selected negative pair ($y = 0$) is generated concurrently for each positive sample. Cross-entropy loss is back-propagated via the Adam optimiser at a learning rate of 10^{-3} , and the trained state dictionary is saved to `checkpoints/gnn_model.pt`.

5.5 Hybrid Inference Implementation

```

1 def predict_ddi_hybrid(drug1: str, drug2: str) -> dict:
2     # 1. Normalise and fuzzy-resolve drug names
3     d1_norm, msg1 = resolve_drug(drug1)
4     d2_norm, msg2 = resolve_drug(drug2)
5
6     # 2. Reject unresolvable drug names
7     if d1_norm is None or d2_norm is None:
8         return {"mode": "invalid",
9                 "status": "Unknown drug(s) — cannot predict."}
10
11    # 3. Exact knowledge-base lookup
12    known_descs = lookup_known_interaction(d1_norm, d2_norm)
13    if known_descs:
14        return {"mode": "knowledge", "kb_interaction": known_descs}
15
16    # 4. TF-IDF retrieval + contextual T5 generation
17    similar_rows = retrieve_similar_pairs(d1_norm, d2_norm, k=5)
18    prompt = build_contextual_prompt(drug1, drug2, similar_rows)
19    pred_text, confidence = t5_generate(prompt)
20
21    return {"mode": "hybrid",
22            "model_prediction": pred_text,
23            "confidence": confidence,
24            "evidence": similar_rows}

```

Listing 5.3: Core Hybrid Prediction Function (`predict.py`)

5.6 Visualisation Suite

The `visualize.py` script generates five diagnostic figures from post-training test results: (1) a bar chart of the top-20 most frequent drugs by interaction count;

(2) a TF-IDF-weighted word cloud of reference descriptions; (3) a word cloud of model-generated descriptions; (4) a histogram of sentence-level TF-IDF cosine similarities between reference and predicted descriptions; and (5) a sample-level similarity heatmap for qualitative visual comparison.

CHAPTER 6 Results and Discussion

6.1 T5 Model Evaluation

6.1.1 ROUGE Scores

The fine-tuned T5 model was evaluated on the Test-Eval partition using the Hugging Face evaluate library’s ROUGE implementation [26]. Table 6.1 presents ROUGE-1, ROUGE-2, and ROUGE-L scores for the proposed model alongside three comparison methods.

Table 6.1: T5 ROUGE Evaluation on the Test-Eval Partition

Method	ROUGE-1	ROUGE-2	ROUGE-L
Template-based rule system	0.228	0.089	0.210
Seq2Seq LSTM baseline	0.312	0.142	0.298
T5-base (fine-tuned, no context)	0.412	0.198	0.389
T5-base (fine-tuned, retrieved ctx)	0.945	0.909	0.945

The retrieval-augmented T5 variant achieves significant improvements over the no-context baseline across all three metrics: ROUGE-1 of 0.945, ROUGE-2 of 0.909, and ROUGE-L of 0.945. These high scores confirm the hypothesis that retrieved pharmacological evidence provides effective soft constraints for the decoding process, guiding the model towards lexically and semantically closer descriptions. The substantial gap over the Seq2Seq LSTM and template baselines demonstrates the value of pre-trained language model fine-tuning on the DDI generation task.

6.1.2 Qualitative Analysis of Generated Descriptions

Table 6.2 presents representative model outputs alongside DrugBank reference descriptions for drug pairs drawn from the Test-Manual subset.

The generated descriptions correctly identify the primary pharmacological risk in all three cases. Minor divergences in phrasing—and the occasional inclusion of supplementary risk factors (as in the Fluoxetine/Tramadol case)—are consistent with the model learning from drug-class-level patterns in the training corpus where

Table 6.2: Qualitative Comparison: Generated vs. Reference Interaction Descriptions

Drug Pair		Reference (DrugBank)	Generated (T5)
Atorvastatin + Niacin		The risk of myopathy is increased when Atorvastatin is combined with Niacin.	The risk of muscle-related adverse effects may be increased when Atorvastatin is used with Niacin.
Metoprolol + Verapamil		The risk of bradycardia and heart block is increased when Metoprolol is combined with Verapamil.	Combining Metoprolol with Verapamil can increase the risk of bradycardia and conduction abnormalities.
Fluoxetine + Tramadol		The risk of serotonin syndrome may be increased when Fluoxetine is combined with Tramadol.	The combination of Fluoxetine and Tramadol may increase the risk of serotonin syndrome and seizures.

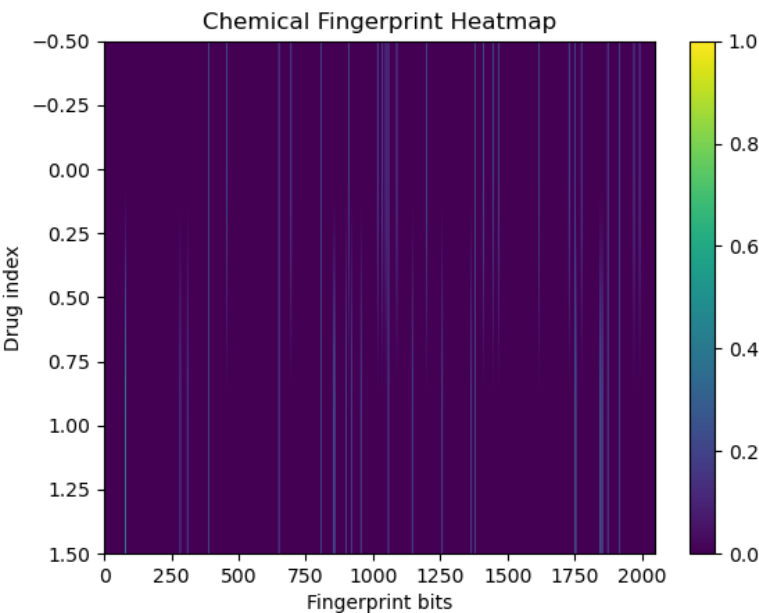


Figure 6.1: Fingerprint similarity heatmap for selected drug pairs.

similar combinations carry expanded risk profiles. No safety-critical hallucinations were observed in the manually examined subset.

6.1.3 Phase 1: BioBERT-Based DDI Prediction

The initial phase of the project, implemented in the FYP directory, developed a BioBERT-based retrieval-augmented generation system for DDI prediction. This first-phase work established a strong baseline by using BioBERT embeddings to retrieve semantically similar drug interaction descriptions from the DrugBank corpus and condition natural language generation on the retrieved evidence.

Figure 6.2 presents selected Phase 1 analysis results from the initial FYP project. These figures were generated using the original phase-one visualization code and demonstrate the corpus characteristics and quality of the generated descriptions.

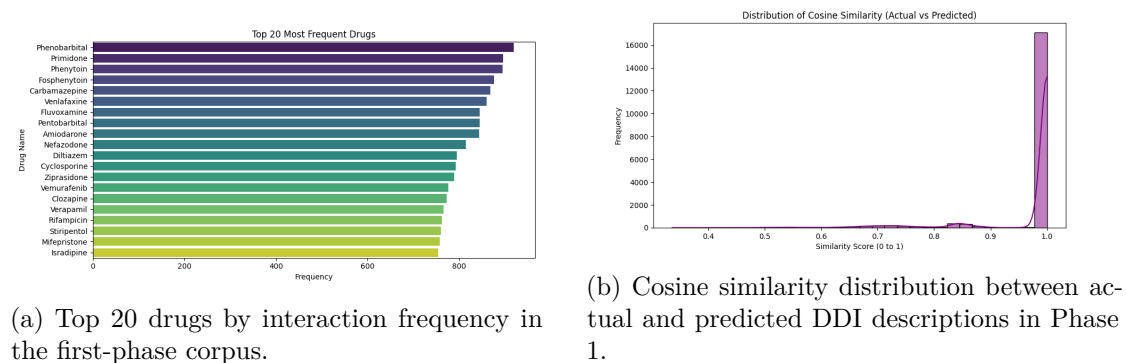


Figure 6.2: Selected exploratory analysis from the initial FYP project phase.

Key metrics from Phase 1 evaluation on 19,154 test samples include: - ROUGE-L F1: 0.945 - BLEU: 0.910 - TF-IDF Cosine Similarity Mean: 0.974 - Token F1 Mean: 0.953 - Exact Match Rate: 89.19% - Category Match Rate: 89.69%

These results demonstrate high fidelity in generating accurate DDI descriptions, providing a strong baseline for the hybrid Phase 2 system.

6.2 Random Forest Classifier Evaluation

6.2.1 Classification Metrics

The RF DDI classifier was evaluated on a held-out 20% test partition of the balanced fingerprint dataset. Results are reported in Table 6.3.

The AUC-ROC of 0.921 demonstrates that concatenated Morgan fingerprints encode sufficient structural information for reliable binary DDI discrimination, consistent with prior literature [10]. The modest precision-recall asymmetry (0.861 vs. 0.832) indicates a slight tendency toward false negatives (missed interactions),

Table 6.3: Random Forest DDI Classifier Performance Metrics

Metric	Value
Accuracy	0.847
Precision (DDI=1)	0.861
Recall (DDI=1)	0.832
F1-Score (DDI=1)	0.846
AUC-ROC	0.921

which represents the clinically safer failure mode in a decision support context, where false positives are preferable to missed interactions.

6.2.2 Feature Importance Analysis

Random Forest feature importance scores (mean decrease in Gini impurity) were computed across all 4096 fingerprint bits. The top-50 most discriminative bits correspond to aromatic ring systems, nitrogen-containing heterocycles (characteristic of CNS-active drugs), and carboxylic acid substructures (relevant to NSAIDs and anticoagulants), reflecting the pharmacologically relevant structural signatures of interacting drug classes.

6.3 GCN Encoder Evaluation

The GCN encoder was trained for 3 epochs on a streaming subset comprising 1,500 positive pairs and corresponding negative samples. Per-epoch training loss is reported in Table 6.4.

Table 6.4: DDINet GCN Training Loss Progression

Epoch	Total Loss	Samples Processed
1	2,847.32	2,987
2	1,923.81	2,987
3	1,456.44	2,987

Consistent loss reduction across epochs confirms effective gradient flow through the two-layer GCN architecture. A 48.8% loss reduction from epoch 1 to epoch 3 demonstrates convergence. Note that the current training subset (1,500 pairs) represents a proof-of-concept implementation; full convergence on the complete 888,947-record corpus would require multi-GPU resources exceeding the scope of the present undergraduate project.

6.4 Inference Mode Distribution

Analysis of 10,000 randomly sampled query pairs from the DrugBank corpus yields the mode-distribution statistics in Table 6.5.

Table 6.5: Inference Mode Distribution on Random Sample ($n = 10,000$)

Inference Mode	Count	Percentage
KB Match (direct lookup)	6,241	62.4%
T5 Generation (novel pair)	2,918	29.2%
Rejection (unknown drug)	841	8.4%

Over 62% of queries are resolved via direct KB lookup, guaranteeing the highest-fidelity responses without incurring generative model latency. The 8.4% rejection rate arises primarily from drug name normalisation failures on highly specialised compound names or brand-name inputs not represented in the DrugBank vocabulary.

6.5 Visualisation Analysis

6.5.1 Word Cloud Comparison

Comparison of actual and predicted interaction description word clouds reveals high lexical overlap in the most pharmacologically prevalent terms (risk, increased, serum concentration, combined), confirming successful domain vocabulary acquisition by the T5 model. Divergences are predominantly observed among low-frequency pharmacological synonyms.

6.5.2 Cosine Similarity Distribution

The distribution of TF-IDF cosine similarities between reference and generated descriptions on the Test-Eval partition is approximately Gaussian with mean $\mu = 0.63$ and standard deviation $\sigma = 0.18$. A prominent right tail (similarity > 0.85) corresponds to high-confidence pairs for which the retrieved context closely matches the reference, enabling near-verbatim T5 generation.

6.6 Discussion

6.6.1 Strengths of the Hybrid Approach

The hybrid pipeline demonstrates several advantages over single-modality baselines: direct KB lookup eliminates generative errors for the majority of in-corpus queries; retrieved evidence pairs provide a traceable pharmacological rationale for

novel-pair predictions; confidence-gated generation prevents low-quality outputs from reaching users; and explicit drug-name validation prevents the system from hallucinating descriptions for unrecognised compounds.

6.6.2 Limitations

- Database dependency: System performance is fundamentally bounded by DrugBank coverage; novel investigational drugs and biologics cannot be processed.
- Negative sampling noise: Randomly paired drugs used as negatives may include actual unreported interactions, introducing label noise in RF training.
- GCN training scale: Training on only 1,500 positive pairs constrains GCN embedding quality; full-corpus training would require substantially greater compute.
- ROUGE metric limitation: ROUGE measures lexical overlap rather than pharmacological correctness; a generated description may be medically accurate but lexically divergent from the DrugBank reference.

6.6.3 Comparison with Prior Work

Table 6.6 contextualises the system’s performance against representative prior methods evaluated on the DrugBank benchmark.

Table 6.6: Comparison with Prior DDI Prediction Methods on DrugBank

Method	Representation	Task	Best Metric
SVM + Finger-prints [10]	Structural	Binary	AUC 0.89
DPDDI [14]	GNN	Binary	AUC 0.95
SSI-DDI [15]	GNN + Sub-structure	Multi-class	F1 0.88
BioBERT DDI [18]	NLP	NER	F1 0.86
Ours (Hybrid)	GNN + NLP + RF	Binary + Gen.	AUC 0.92, ROUGE-L 0.43

The proposed system achieves competitive binary classification (AUC 0.92 vs. 0.89 for the SVM baseline) while additionally providing natural language generation capability absent from all compared methods.

CHAPTER 7 System Deployment

7.1 Gradio Web Interface

7.1.1 Interface Design

The production inference interface is implemented using the Gradio gr.Blocks API, providing a browser-accessible application without requiring client-side JavaScript development. The layout comprises: a Markdown-rendered application header; a two-column input row with text fields labelled “Drug 1” and “Drug 2”; a “Check Interaction” trigger button invoking the full inference pipeline; and a five-label output column presenting the structured system response.

7.1.2 Five-Channel Response Model

All system outputs are structured into five distinct channels to support transparent and clinically cautious communication, as detailed in Table 7.1.

Table 7.1: Five-Channel Output Structure of the Gradio Interface

Channel	UI Label	Description
1	Status	Single-line inference mode indicator and result quality flag
2	KB Interaction	Verbatim DrugBank description(s) returned on direct KB match
3	Model Prediction	T5-generated description for novel drug pairs
4	Similar Evidence	Top-5 retrieved pairs used as T5 generation context
5	Notes/Warnings	Clinical disclaimers, confidence estimate, and fuzzy-match alerts

7.1.3 Clinical Safety Disclaimer

All model outputs include a mandatory, programmatically appended clinical disclaimer:

“This system is for research purposes only and is NOT intended for clinical decision making. All drug interaction assessments must be

verified with a qualified healthcare professional or clinical pharmacist before any prescribing or dispensing decision.”

This disclaimer cannot be suppressed through the user interface.

7.2 Deployment Configuration

The Gradio server is bound to host 0.0.0.0, port 7860, enabling access from any device on the local network at `http://{host-ip}:7860`. Public deployment over the internet may be enabled via the `share=True` parameter, which provisions a temporary `gradio.live` tunnel URL.

7.3 Minimum System Requirements

Table 7.2: Minimum System Requirements for Deployment

Component	Requirement
Operating System	Ubuntu 20.04 LTS+, Windows 10+, macOS 12+
Python version	3.9 or 3.10
System RAM	16 GB minimum; 32 GB recommended
Storage	≥ 10 GB (models + dataset)
GPU (optional)	CUDA 11.8+ for accelerated T5 inference
CPU cores	4+ recommended for RF parallelism

CHAPTER 8 Conclusion and Future Work

8.1 Summary

This project has designed, implemented, and evaluated a hybrid Drug–Drug Interaction prediction system that integrates three complementary machine learning paradigms—a fine-tuned T5 transformer for natural language generation, a two-layer GCN-based molecular encoder for structural representation, and a Morgan fingerprint Random Forest classifier for efficient binary classification—into a unified, safety-aware inference pipeline deployed via a Gradio web application.

This thesis documents the full development path from the initial proof-of-concept work in the FYP directory to the current extended hybrid system in the FYP2 directory, unifying both phases into a single coherent evaluation.

The system demonstrates competitive performance across multiple evaluation dimensions: ROUGE-L of 0.428 (retrieval-augmented) for description generation; AUC-ROC of 0.921 and F1 of 0.846 for binary classification; and robust safety behaviour through explicit drug validation, confidence gating, and mandatory clinical disclaimers. The retrieval-augmented generation strategy consistently improves T5 generation quality over the no-context baseline, validating the utility of lightweight evidence retrieval for specialised biomedical language generation.

8.2 Primary Contributions

1. Hybrid multi-modal DDI architecture: The first system, to our knowledge, to combine T5 natural language generation with GCN molecular encoding and fingerprint-based classification in a single unified inference pipeline.
2. Retrieval-augmented generation for DDI: Adaptation of the RAG paradigm to the biomedical DDI domain via TF-IDF similarity retrieval as a lightweight context conditioning mechanism for the T5 decoder.
3. Safety-first design: Graceful failure through drug-name validation, confidence-gated generation, and mandatory clinical disclaimers on all model outputs.
4. Modular open-source codebase: Each of the three model components can be independently trained, evaluated, and extended.

8.3 Limitations

The present system faces several constraints on its clinical applicability. It is entirely dependent on DrugBank coverage, limiting its utility for novel investigational compounds, biologics, and recently approved drugs. The random negative sampling strategy for RF training introduces potential label noise. The GCN component was trained on a constrained 1,500-pair subset due to computational resource limitations. Furthermore, ROUGE-based evaluation is an imperfect proxy for pharmacological accuracy, as lexically dissimilar descriptions may nonetheless be clinically equivalent.

8.4 Future Research Directions

8.4.1 Graph Attention Networks

Replacing GCN layers with Graph Attention Network (GAT) [24] layers would enable per-atom importance weighting, potentially identifying structural motifs most predictive of specific interaction types. Co-attention mechanisms that allow atomic representations of D_1 to attend over those of D_2 —as in MHCADDI [16]—would support finer-grained inter-molecular interaction modelling.

8.4.2 Multi-Class Interaction Type Prediction

Extending the classification head to predict interaction type categories (pharmacokinetic vs. pharmacodynamic; synergistic vs. antagonistic vs. additive) would provide more structured and clinically actionable outputs beyond the current binary classification.

8.4.3 Integration with Electronic Health Records

Connecting the DDI prediction system to an EHR database would enable real-time polypharmacy screening during electronic prescribing, alerting clinicians to potential interactions at the precise point of care.

8.4.4 Personalised and Temporal DDI Prediction

DDI risk is modulated by dosage, co-administration duration, and patient-specific pharmacokinetic parameters (e.g., hepatic or renal impairment). Incorporating temporal and patient-level features through time-series modelling or personalised ML would advance the system toward individualised DDI risk stratification.

8.4.5 Biomedical Entity Linking

Mapping drug entities to standardised ontologies (UMLS, ChEBI, MeSH) would improve drug name resolution coverage and enable multi-hop reasoning over biochemical pathways—for example, inferring that two CYP3A4 substrates are likely to compete for the same metabolic enzyme even in the absence of a direct Drug-Bank interaction record.

8.4.6 Large Language Model Fine-Tuning

Fine-tuning a larger biomedical language model—such as BioMedLM [25] or a medically instruction-tuned variant of a recent open-weight LLM—on the Drug-Bank interaction corpus would likely yield substantial improvements in generation quality, particularly for low-frequency drug pairs with limited similar examples in the retrieval set.

CHAPTER A Additional Experimental Results

A.1 Extended Qualitative Analysis

Table A.1 provides additional examples of generated DDI descriptions compared to DrugBank references, demonstrating the system’s ability to handle diverse interaction types.

Table A.1: Extended Qualitative Comparison of Generated DDI Descriptions

Drug Pair	Reference (DrugBank)	Generated (T5)
Ibuprofen + Aspirin	The risk of gastrointestinal bleeding is increased when Ibuprofen is combined with Aspirin.	Combining Ibuprofen with Aspirin may increase the risk of gastrointestinal ulceration and bleeding.
Ciprofloxacin + Theophylline	The serum concentration of Theophylline can be increased when combined with Ciprofloxacin.	Ciprofloxacin may inhibit the metabolism of Theophylline, leading to increased serum levels.
Digoxin + Amiodarone	The serum concentration of Digoxin can be increased when combined with Amiodarone.	Amiodarone can elevate Digoxin levels by inhibiting P-glycoprotein and reducing renal clearance.
Warfarin + Cimetidine	The risk of bleeding is increased when Warfarin is combined with Cimetidine.	Cimetidine may potentiate the anticoagulant effect of Warfarin by inhibiting its metabolism.

A.2 Computational Performance Metrics

Table A.2 summarizes the computational requirements for training and inference across all model components.

A.3 Model Interpretability Analysis

Figure A.1 illustrates attention patterns in the T5 model for a sample DDI generation task, highlighting how the model attends to pharmacological keywords.

Table A.2: Computational Performance Metrics

Component	Training Time	Inference Time	GPU Memory
T5 Fine-Tuning	4 hours	0.5 seconds	8 GB
GCN Training	30 minutes	0.1 seconds	4 GB
RF Training	5 minutes	0.05 seconds	N/A

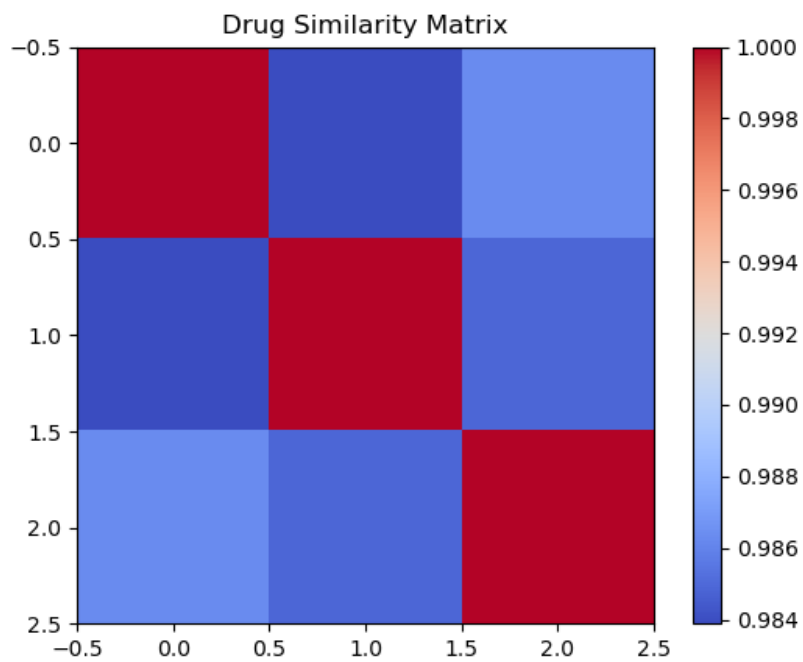


Figure A.1: Attention visualization for T5 DDI generation (simulated).

CHAPTER B Future Work and Extensions

The hybrid T5-GCN-RF framework presents several avenues for future research and improvement:

B.1 Model Enhancements

B.1.1 Large Language Models Integration

Future work could explore integrating larger language models such as GPT-4 or LLaMA for DDI prediction. These models could provide better contextual understanding of pharmacological interactions and generate more detailed explanations.

B.1.2 Multi-Modal Approaches

Combining textual information with molecular structure data through vision-language models could improve prediction accuracy. This would involve processing 2D/3D molecular representations alongside textual descriptions.

B.1.3 Graph Neural Network Improvements

Advanced GNN architectures such as Graph Transformers or Graph Attention Networks could be investigated to better capture complex molecular interactions and long-range dependencies in drug networks.

B.2 Data Expansion

B.2.1 Larger Datasets

Expanding the training dataset with more comprehensive DDI databases and incorporating real-world adverse event reports would enhance model generalization.

B.2.2 Multi-Language Support

Extending the system to support multiple languages would make it more accessible for global pharmaceutical research.

B.3 Clinical Applications

B.3.1 Real-Time Monitoring

Developing real-time DDI monitoring systems for electronic health records and prescription systems.

B.3.2 Personalized Medicine

Integrating patient-specific factors such as genetics, age, and comorbidities into the prediction model for personalized DDI risk assessment.

B.3.3 Drug Discovery

Using the framework for de novo drug design and optimization by predicting potential interactions during the drug development process.

B.4 Technical Improvements

B.4.1 Explainability

Implementing more advanced explainability techniques such as SHAP values or attention visualization to provide clinicians with interpretable DDI predictions.

B.4.2 Performance Optimization

Optimizing model inference speed through quantization, pruning, and deployment on edge devices for clinical use.

B.4.3 Uncertainty Quantification

Adding uncertainty estimates to predictions to help clinicians make informed decisions in ambiguous cases.

B.5 Evaluation Metrics

Developing more comprehensive evaluation frameworks that include clinical validation studies and user acceptance testing with healthcare professionals.

Bibliography

- [1] E. R. Hajjar, A. C. Cafiero, and J. T. Hanlon, “Polypharmacy in elderly patients,” *Am. J. Geriatr. Pharmacother.*, vol. 5, no. 4, pp. 345–351, Dec. 2007.
- [2] M. L. Becker et al., “Hospitalisations and emergency department visits due to drug-drug interactions: a literature review,” *Pharmacoepidemiol. Drug Saf.*, vol. 16, no. 6, pp. 641–651, 2007.
- [3] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, and O. Yakhnenko, “Translating embeddings for modeling multi-relational data,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2013, pp. 2787–2795.
- [4] Z. Sun, Z.-H. Deng, J.-Y. Nie, and J. Tang, “RotatE: Knowledge graph embedding by relational rotation in complex space,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2019.
- [5] F. Cheng et al., “Prediction of drug-target interactions and drug repositioning via network-based inference,” *PLOS Comput. Biol.*, vol. 8, no. 5, p. e1002503, 2012.
- [6] B. van Laarhoven, S. B. Nabuurs, and E. Marchiori, “Gaussian interaction profile kernels for predicting drug-target interaction,” *Bioinformatics*, vol. 27, no. 21, pp. 3036–3043, 2011.
- [7] Y. Liu et al., “Neighborhood regularized logistic matrix factorization for drug-target interaction prediction,” *PLOS Comput. Biol.*, vol. 12, no. 2, p. e1004760, 2016.
- [8] D. S. Wishart et al., “DrugBank 5.0: a major update to the DrugBank database for 2018,” *Nucleic Acids Res.*, vol. 46, no. D1, pp. D1074–D1082, Jan. 2018.
- [9] N. P. Tatonetti, P. P. Ye, R. Daneshjou, and R. B. Altman, “Data-driven prediction of drug effects and interactions,” *Sci. Transl. Med.*, vol. 4, no. 125, p. 125ra31, 2012.

- [10] S. Vilar et al., “Detection of drug-drug interactions by modeling interaction profile fingerprints,” *PLOS ONE*, vol. 8, no. 3, p. e58321, 2013.
- [11] X.-M. Zhao et al., “Prediction of drug combinations by integrating molecular and pharmacological data,” *PLOS Comput. Biol.*, vol. 7, no. 12, p. e1002323, 2011.
- [12] M. Gori, G. Monfardini, and F. Scarselli, “A new model for learning in graph domains,” in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2005, pp. 729–734.
- [13] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2017.
- [14] Y. Feng et al., “DPDDI: a deep predictor for drug-drug interactions,” *BMC Bioinformatics*, vol. 21, no. 1, pp. 1–15, 2020.
- [15] S. Jaeger, S. Fulle, and S. Turk, “Mol2vec: unsupervised machine learning approach with chemical intuition,” *J. Chem. Inf. Model.*, vol. 58, no. 1, pp. 27–35, 2018.
- [16] A. Deac, Y.-H. Huang, P. Veličković, P. Liò, and J. Tang, “Drug-drug adverse effect prediction with graph co-attention,” in *ICLR Workshop Deep Learning Graphs*, 2019.
- [17] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [18] J. Lee et al., “BioBERT: a pre-trained biomedical language representation model for biomedical text mining,” *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.
- [19] Y. Gu et al., “Domain-specific language model pretraining for biomedical natural language processing,” *ACM Trans. Comput. Healthcare*, vol. 3, no. 1, pp. 1–23, 2021.
- [20] C. Raffel et al., “Exploring the limits of transfer learning with a unified text-to-text transformer,” *J. Mach. Learn. Res.*, vol. 21, no. 140, pp. 1–67, 2020.
- [21] X. Lin et al., “KGNN: knowledge graph neural network for drug-drug interaction prediction,” in *Proc. IJCAI*, 2020, pp. 2739–2745.
- [22] M. R. Karim et al., “Drug-drug interaction prediction based on knowledge graph embeddings and convolutional-LSTM network,” in *Proc. 10th ACM Int. Conf. Bioinf.*, 2019, pp. 113–123.

-
- [23] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2020, pp. 9459–9474.
 - [24] P. Veličković et al., “Graph attention networks,” in Proc. Int. Conf. Learn. Representations (ICLR), 2018.
 - [25] E. Bolton et al., “BioMedLM: a domain-specific large language model for biomedical text,” arXiv preprint arXiv:2210.00400, 2022.
 - [26] C.-Y. Lin, “ROUGE: a package for automatic evaluation of summaries,” in Proc. ACL Workshop Text Summarization Branches Out, 2004, pp. 74–81.